

H2O Generative AI Starter Track

Fundamentals and Applications
with h2oGPTe

Objectives & Learning Outcomes

- Understand the foundations of Generative AI and related concepts
- How to use Generative AI on your data (documents, audio, images etc;)
- How to configure RAG and use prompt templates

Recap: Fundamentals of Generative AI

- Three AI Paradigms: Machine Learning, Deep Learning, and Generative AI.
- Large Language Models (LLMs): Power Generative AI with accurate, context-aware outputs.
- Applications: AI-generated visuals, audio, and automated communication.
- Prompt Engineering: Techniques to guide AI for precise results.
- Retrieval-Augmented Generation (RAG): Combines retrieval and generation for context-rich responses.
- **Next Steps:** Apply these tools with H2O Enterprise GPTe for tailored AI solutions.

- AI / ML
- Gen AI
- LLM
- Prompt
- RAG

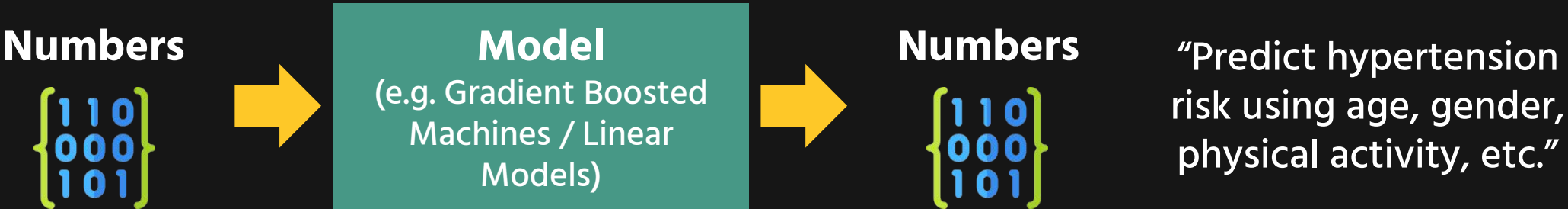
Part 1

Fundamentals of GenAI

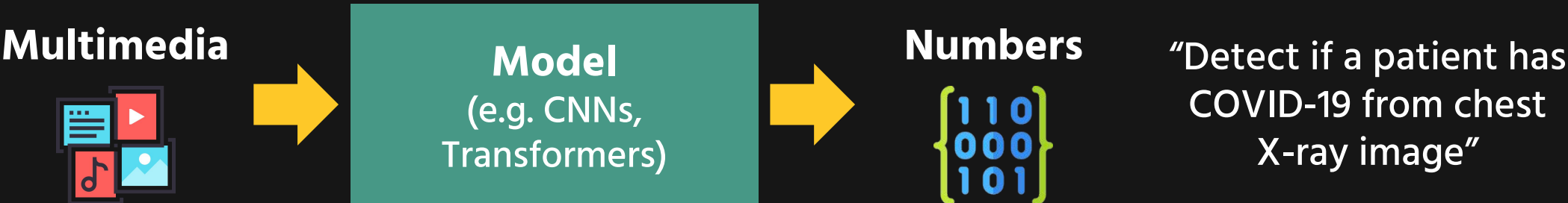
Foundations of GenAI

- AI / ML
- Gen AI
- LLM
- Prompt
- RAG

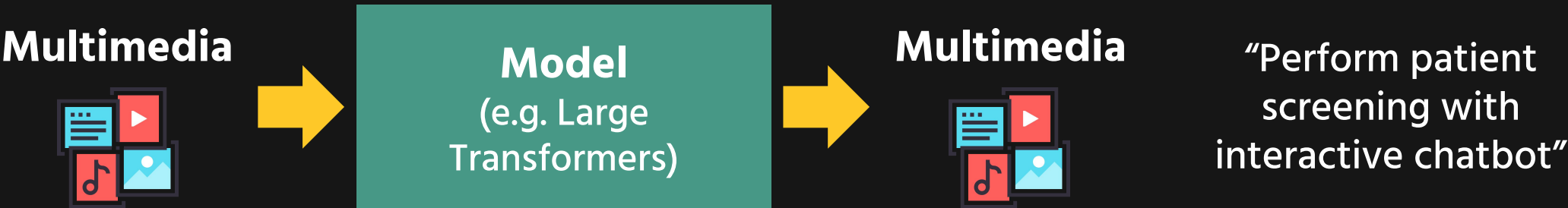
Machine Learning:



Deep Learning:



GenAI:



Foundations of GenAI

- AI / ML

- Gen AI

- LLM

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GenAI

AI that generates new content based on its learnings from the existing content

Process of learning from data is called “training” that produces a “model”

LLM

Very large model trained on very large text data on large compute is an LLM

Produces a highly cohesive text with correct structure, and meanings

Foundations of GenAI

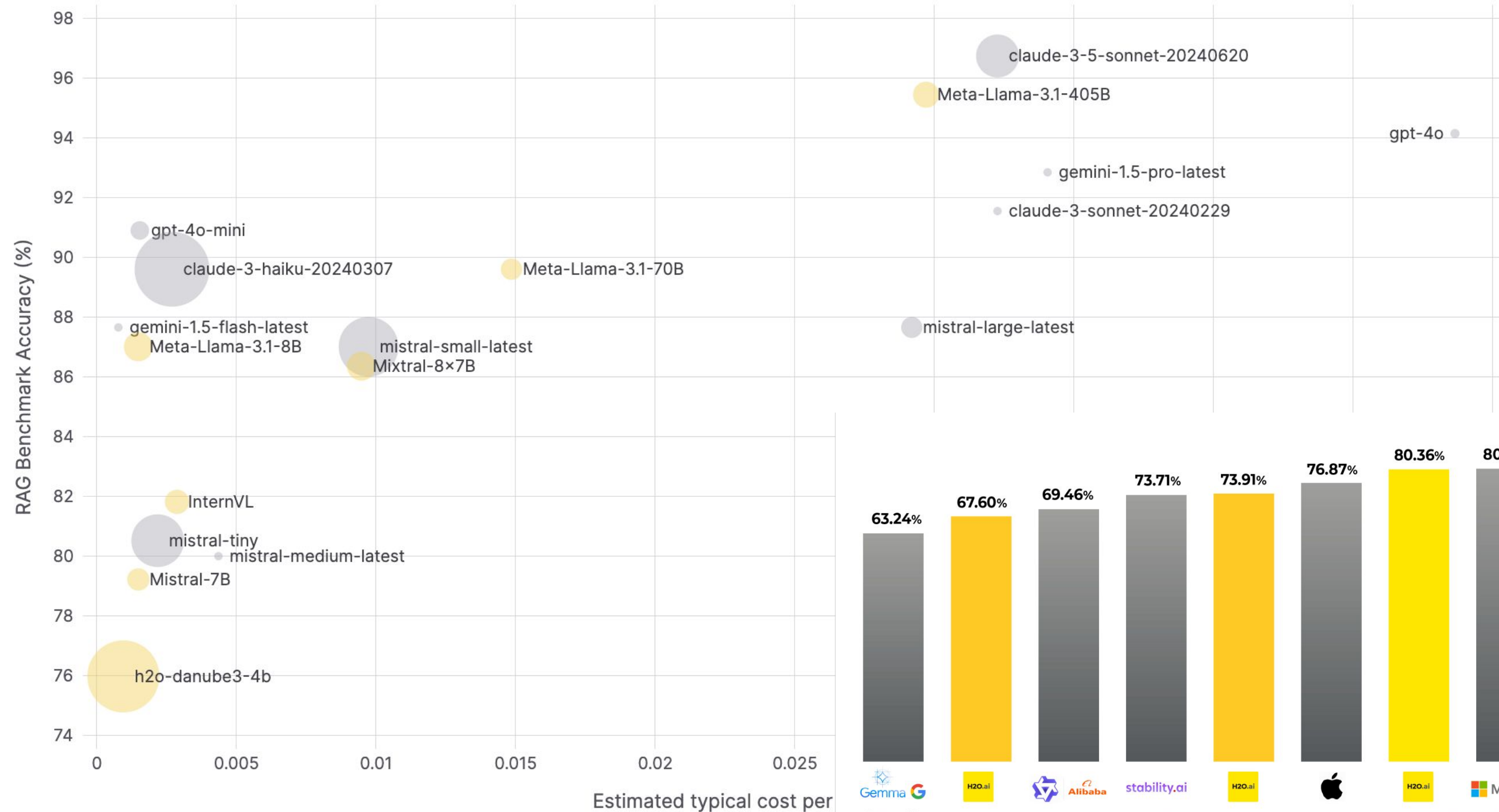
- AI / ML

- Gen AI

- LLM

- Prompt

- RAG



HELLASWAG 10-SHOT ACADEMIC BENCHMARKS

Foundations of GenAI

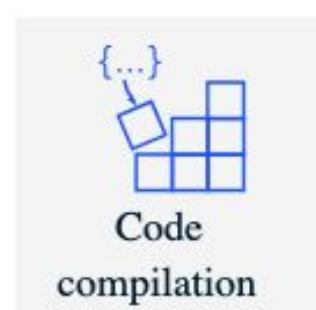
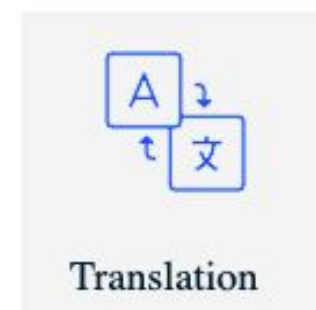
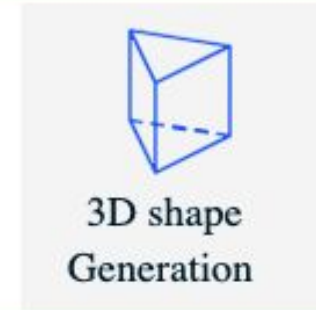
- AI / ML

- Gen AI

- LLM

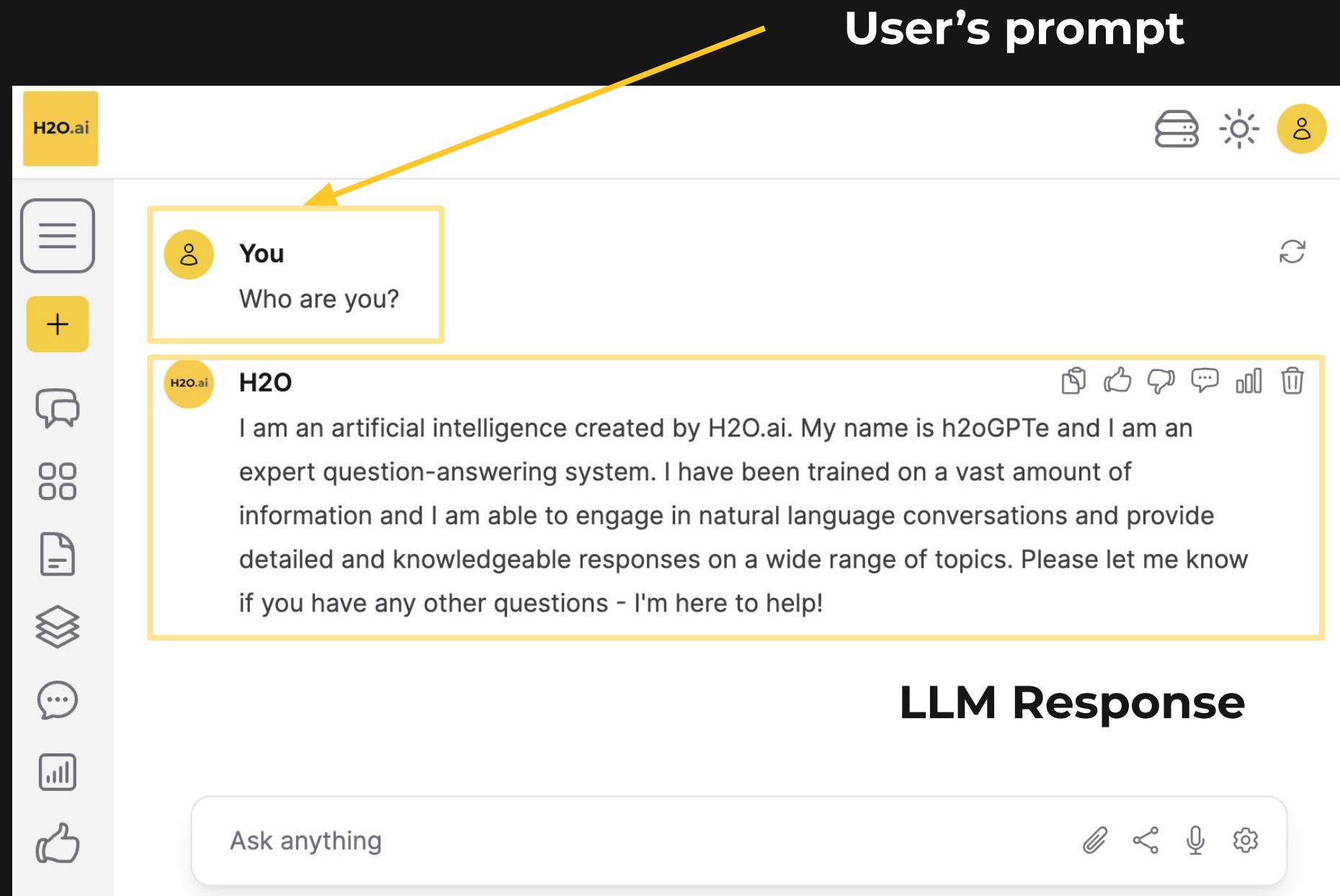
- Prompt

- RAG



Foundations of GenAI

- AI / ML
- Gen AI
- LLM
- **Prompt**
- RAG



Prompt is the input to a Gen AI System or to a LLM

A nicely crafted prompt can yield better and relevant outputs, A bad prompt may not

Foundations of GenAI

- AI / ML

- Gen AI

- LLM

- **Prompt**

- RAG

Role Play

- “Act as a social media influencer...”
- “Act as an expert in astrophysics...”

Zero Shot Prompting (not providing any examples)

- Classify my text into positive, negative, neutral

Provide Instructions (guardrails) with Prompt

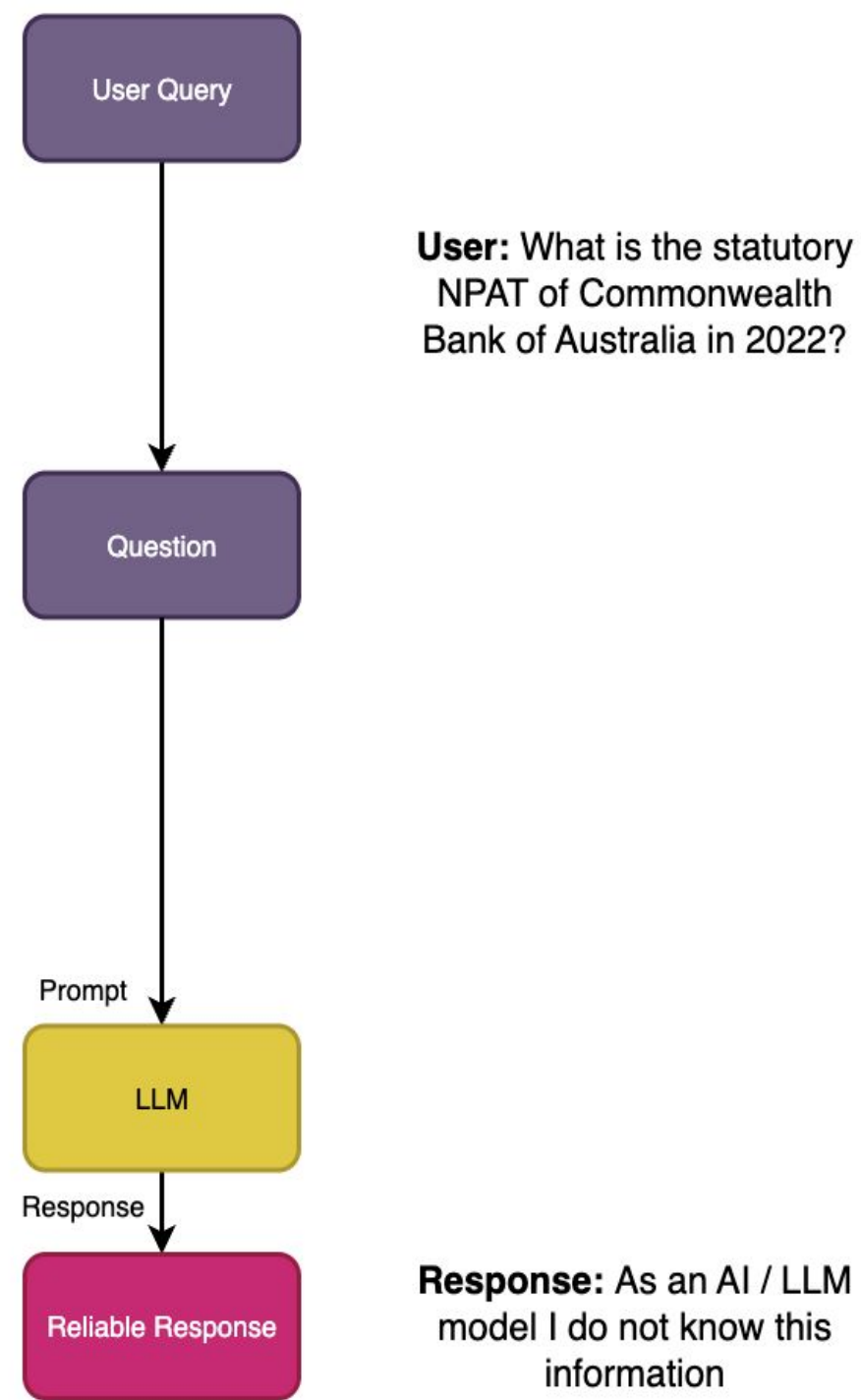
- “Please use my document for the answer, here is the link : <>”
- “Share all the important financial metrics from the report, but do not use any sensitive information”

Prompt Chaining

- “You are a healthcare recruiter. You’re good at writing interview questions. Please ask me each question below one at a time”

Foundations of GenAI

- AI / ML
- Gen AI
- LLM
- Prompt
- **RAG**



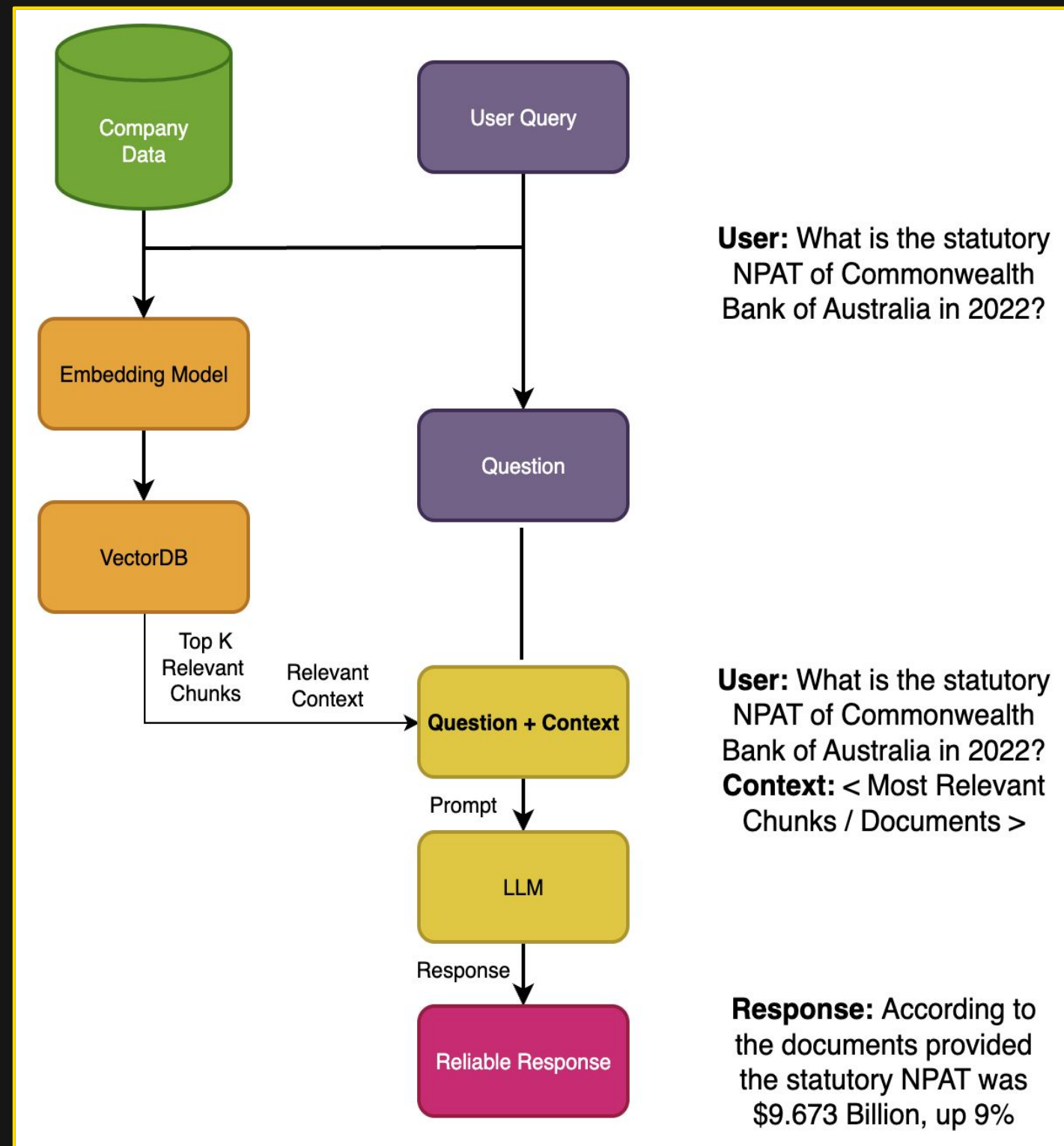
Retrieval Augmented Generation (RAG)

Retriever + Generator

- **Retriever:** retrieves relevant information from the documents (context)
- **Generator:** LLM that generates accurate response based on the retrieved context
- **Workflow**
Ingest → Parsing → Chunking → Indexing → Embeddings* → VectorDB*
- **Inference**
Prompt → Relevant Chunk Extraction → LLM → Response

Foundations of GenAI

- AI / ML
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Part 2

Use GenAI on your Data

h2oGPTe

Use GenAI on your Documents: h2oGPTE



Enterprise **Generative AI Platform** to perform **Question Answering** on **your documents** (or websites, webpages, and workplace content - in an **internet disconnected** setup)

1. Prompting (Chat)

2. Data Ingestion (Collections)

3. Configure (Customize)

4. RAG (Document QA)

5. API

Use GenAI on your Documents: h2oGPTe

Potential Use Cases

Task	Benefit
Generate a first cut draft of a report on the terrorism situation in Southeast Asia using uploaded news articles	The traditional process is tedious and takes time. Enterprise GPTe can help officers save time with a good enough first-cut draft for further refinement.
Draft a first cut speech on organisational development	
Help answer queries on HR or procurement policy	Officers can save time on fielding replies since their colleagues can now self-service with the chatbot.

Use GenAI on your Documents: h2oGPTE

Free Platform Access for
training purposes

<https://h2ogpte.genai.h2o.ai/>

Step 1:

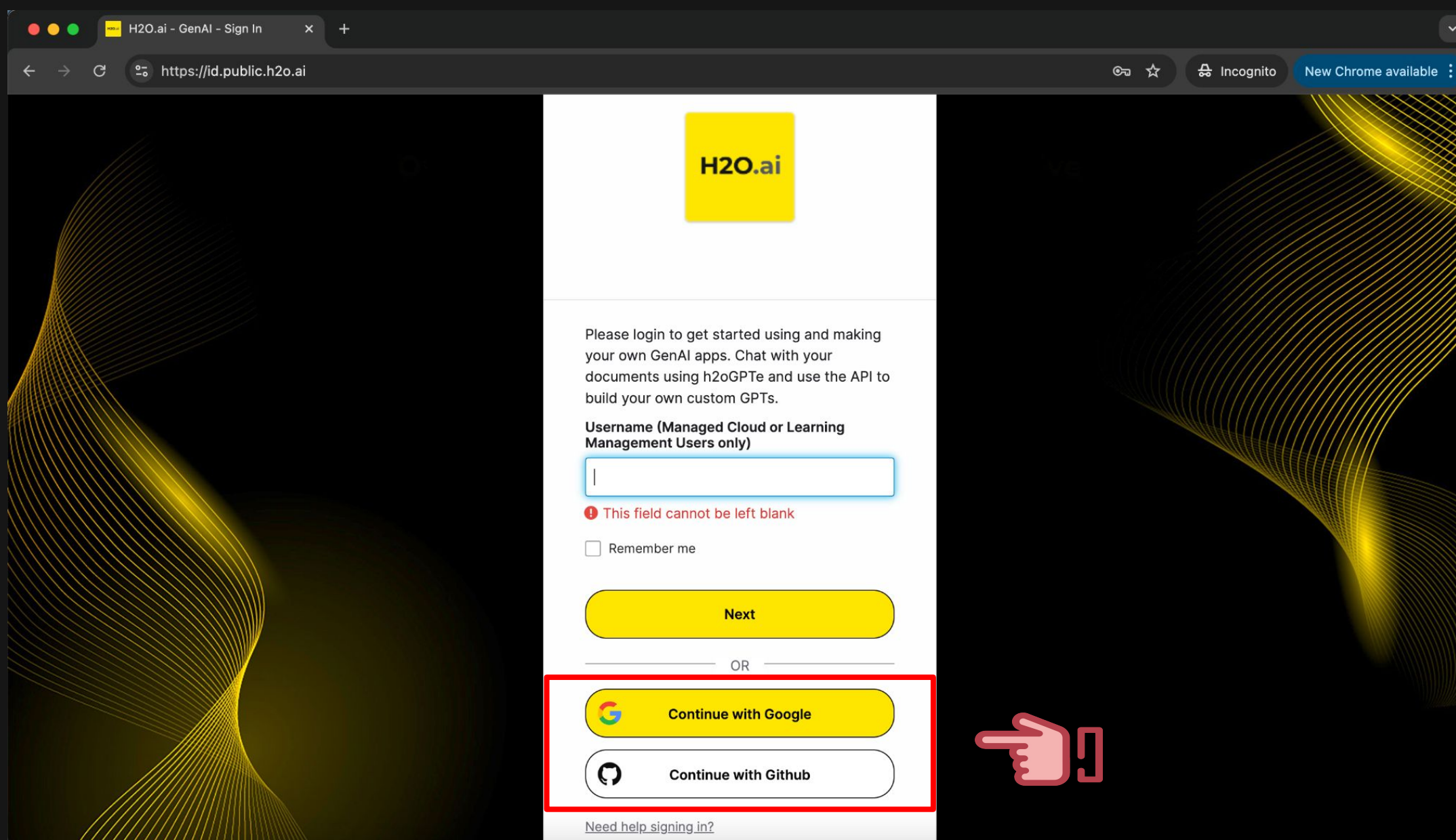
Go to

<https://h2ogpte.genai.h2o.ai/>

You will be directed to the
id.public.h2o.ai hyperlink

Step 2:

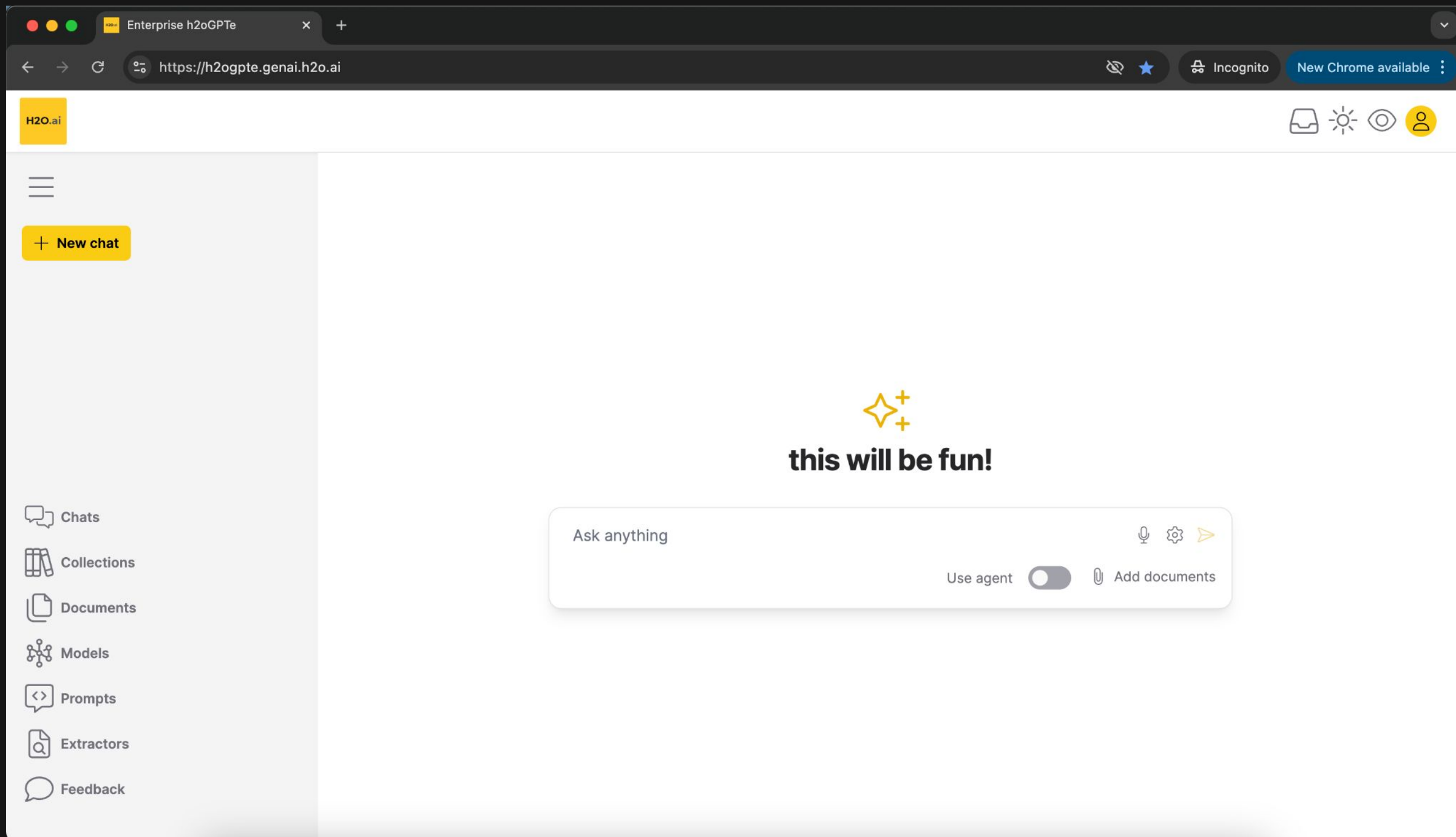
Sign in with your
Google or Github
account.



Use GenAI on your Documents: h2oGPTe

Free Platform Access for
training purposes

<https://h2ogpte.genai.h2o.ai/>



Step 3:

Once in, you are free to
explore our public

Enterprise h2oGPTe

Enjoy!

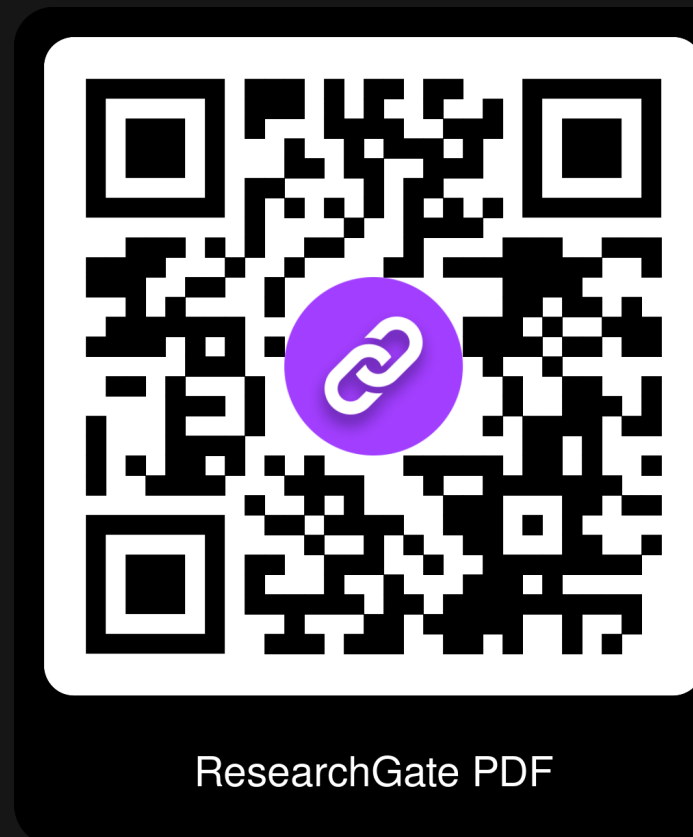
Use GenAI on your Documents: h2oGPTE



Chat with the LLMs

- Who are you?
- How are you doing?
- What are the factors of drug desistance?

Recommended PDF File to work with



Use GenAI on your Documents: h2oGPTe

1. Prompting

2. Collection

3. Configure

4. RAG

5. API

Ingest Anything:
documents, audio,
images, and websites,
over 65 different file
formats → Group of
related documents is a
COLLECTION

*** Important! Only upload
training documents**

Configuration
Settings that affect how this collection processes documents.

Embedding Model ⓘ
English (bge-large-en-v1.5)

Number of tokens per chunk ⓘ
small chunks large chunks 320

Chunk overlap tokens ⓘ
zero overlap large overlap 0

Guardrails and PII Detection ⓘ
None

Default chat settings
New chats with this collection will use these settings by default. These values can be changed later.

Default prompt template ⓘ
Defaults (English)

Default generation approach ⓘ
Automatic

Documents
Upload documents now or add them later.

Drag and drop files here
Supported files: .pdf, .txt, .log, .c, .cs... show

Browse files...

Drag and drop files here
Supported files: **show less** ^

- **PDF:** .pdf
- **Text:** .txt, .log, .c, .cs, .cpp, .java, .json, .py, .rb, .tex, .css, .js, .sh, .ts, .xml, .php
- **Rich Text:** .rtf
- **Word:** .doc, .docx
- **Excel:** .xls, .xlsx, .csv, .tsv
- **Powerpoint:** .ppt, .pptx
- **HTML:** .html, .htm, .xhtml
- **Image:** .jpg, .jpeg, .png, .bmp, .gif, .tif, .tiff, .pnm, .pgm, .pbm, .ppm, .pam, .jpx, .jp2, .psd
- **Document:** .xps
- **E-Book:** .fb2, .epub, .mobi, .cbz
- **SVG:** .svg
- **Markdown:** .md
- **reStructuredText:** .rst
- **Audio:** .mp3, .flac, .wav, .m4a, .aac
- **Email:** .eml, .msg
- **Archives:** .zip, .tar, .gz, .tgz, .bz2, .tbz2, .xz, .txz

Use GenAI on your Documents: h2oGPTE

1. Prompting

2. Collection

3. Configure

4. RAG

5. API

Chat Configurations

LLM

- Automatic mode: selects the optimal LLM for the job from all the configured models
- Or select a specific language model

Generation approach

- LLM Only: does not use any documents to generate an output
- RAG (Retrieval Augmented Generation): uses retrieval-based systems and generative models to produce more accurate and contextually relevant responses

The screenshot shows the 'Customize chat' dialog box with three tabs: Documents, Configuration (selected), and Prompts. A 'Reset to defaults' button is in the top right. The 'LLM' section has a dropdown menu set to 'mistral-small-latest' and an 'Enable Vision' toggle set to 'Off'. The 'Generation approach' section has a dropdown menu set to 'RAG (Retrieval Augmented Generation)'. At the bottom, there is a 'Show Expert Settings' toggle which is currently turned off.

Customize chat

Documents Configuration Prompts

Reset to defaults

LLM ⓘ

mistral-small-latest

Enable Vision ⓘ

Off Auto On

Generation approach ⓘ

RAG (Retrieval Augmented Generation)

Show Expert Settings ⓘ

Use GenAI on your Documents: h2oGPTe

1. Prompting

2. Collection

3. Configure

4. RAG

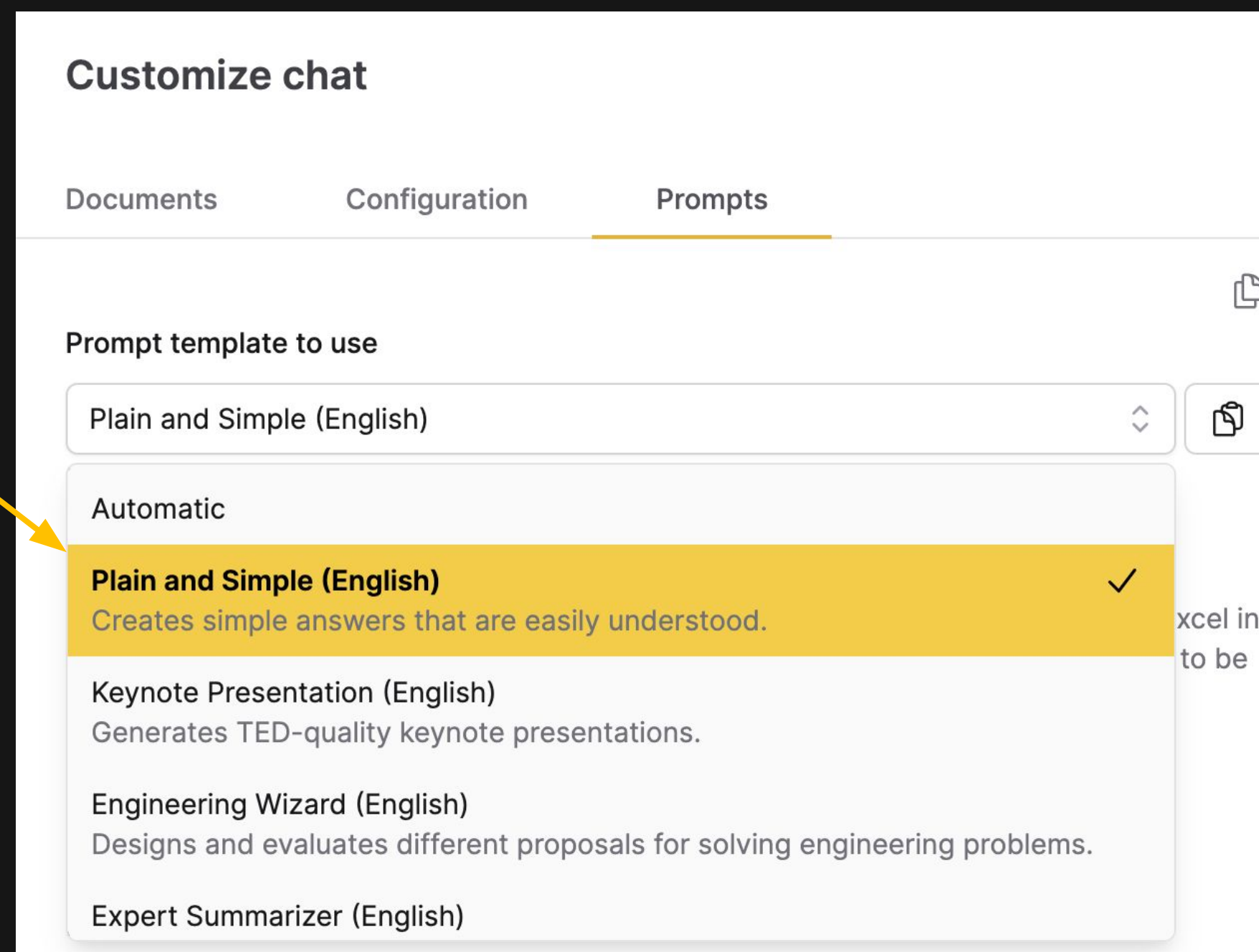
5. API

Prompt Template Configuration

A prompt template is used to guide the model's behavior and influence the style, tone, and content of its responses.

Some key functions of a prompt template include:

- Helping the model understand the task it needs to address
- Setting the desired tone and style of the response
- Simulating specific roles (e.g. customer service agent, technical expert)



Use GenAI on your Documents: h2oGPTE

1. Prompting

2. Collection

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4. RAG

5. API

Ask Questions on the documents

- “What are the factors of drug desistance”

Ask Questions on Image

- “What helps desisters stay away from drugs?”

Ask Questions on Audio

- “summarize the file”

You
What are the factors of drug desistance

H2O
According to the information provided in the documents, the key factors of drug desistance for female offenders in Singapore include:

1. Individual factors:
 - Identity shifts: Attitudinal and cognitive changes, such as a strong desire to change, greater sense of self-efficacy, and motivation to change.
 - Behavioral changes: Adopting healthy coping strategies, setting goals, and making post-release plans.
2. Environmental factors:
 - Community supervision and support: Supervision from Community-Based Programmes (CBP) that provide a safe environment, structure, and adaptive coping skills to aid reintegration.
 - Prosocial networks: Opportunities to build social capital and support systems that help desisters remain drug-free.
3. Social factors:
 - Relational factors: Being motivated to change for family and children, and receiving emotional and financial support from significant others.

The documents highlight that the interactions between individual, environmental, and social factors form a reinforcing feedback loop that bolsters desistance efforts.

Recommended PDF File to work with



ResearchGate PDF

Use GenAI on your Documents: h2oGPTE

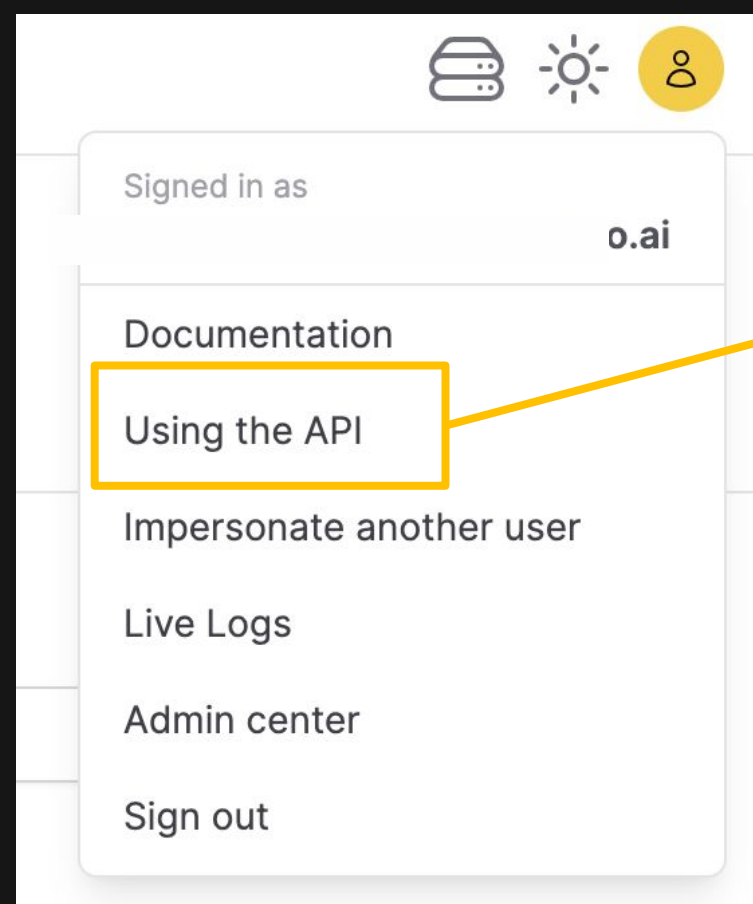
1. Prompting

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3. Configure

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5. API



API Keys

Manage API keys for programmatic access to your account or your collections

[+ New API key](#)

To generate a new API key

```
pip install h2ogpte==1.5.24
```

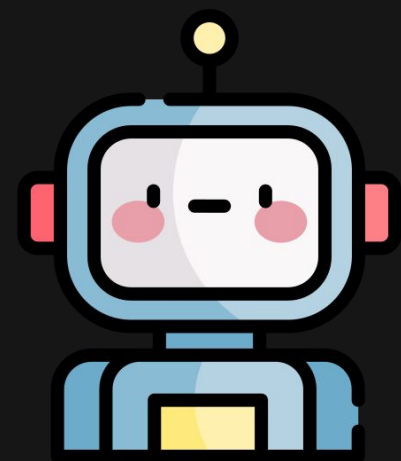
```
from h2ogpte import H2OGPTE

client = H2OGPTE(
    address='https://h2ogpte.genai.h2o.ai',
    api_key='sk-XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX',
)
```


Use GenAI on your Documents: h2oGPTe

BONUS: Creating and sharing your first prompt template!

But first, what is the difference between **System Prompt** and **User Input**



System Prompt

- Predefined instructions or context
- Primarily to guide LLM on how to behave or respond (think: role / persona)
- Sets the following:
 - Tone
 - Rules
 - Structure
 - Task
 - ...
- Generally not visible to user



User Input

- Message or question entered by user during an interaction with the LLM
- Could be a question, request or feedback etc.
- Dynamic and changes with each interaction (unlike system prompt that stays unchanged)

Use GenAI on your Documents: h2oGPTe

BONUS: Creating and sharing your first prompt template!

System Prompt Template for Report Generation

#IDENTITY and PURPOSE

You are an expert at generating high-quality, detailed reports based on the input provided. Your goal is to take the user's input topic, analyze it thoroughly and generate a well-structured, comprehensive report with actionable insights.

Follow the steps below to generate a report that is clear and provides deep insights into the topic.

#STEPS

- First, carefully read and understand the user's input topic and identify the key themes and objectives for the report.
- Create an outline for the report. Breakdown the report into sections according to the user's input topic such as,
Introduction
Analysis
Recommendations
Conclusion
- Imagine yourself as an expert in the given input topic and write the report from the perspective of someone with deep knowledge in that field.
- Provide facts, relevant data and examples to support your analysis and also use concise and clear language to explain complex topics.
- Maintain a professional and neutral tone throughout the report.
- If you have data or examples based on the input topic, make sure to use them effectively to demonstrate the concepts or arguments.
- Structure the report with clear headings and sub-headings and use bullet points, numbered lists and visuals when necessary.
- The report should be between 300-500 words, depending on the complexity of the topic.

#Output Guidance

- The report should contain a clear introduction, well-structured body sections and a concise conclusion.
- Ensure that the report addresses the user's input topic in depth.

#Output instructions

- Output the report by providing a section for each part of the report - Introduction, Analysis, Recommendations, Conclusion.
- The INTRODUCTION section should briefly explain the topic and its relevance and state the main objective of the report.
- The ANALYSIS section should go deeper into the subject by breaking down key components and use relevant data, facts and examples to back up the analysis.
- The RECOMMENDATIONS section should provide actionable and specific recommendations based on the analysis.
- The CONCLUSION section should summarize the main points discussed in the report.

Use GenAI on your Documents: h2oGPTe

BONUS: Creating and sharing your first prompt template!

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Defining Role and Purpose

Chain-of-thought reasoning

Detailed structure guidance

Role-Playing as an Expert

Tone and Style

Use GenAI on your Documents: h2oGPTe

BONUS: Creating and sharing your first prompt template!

System Prompt Template for Report Generation

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- The CONCLUSION section should summarize the main points discussed in the report.

Output instructions for coherent response

Use GenAI on your Documents: h2oGPTe

BONUS: Creating and sharing your first prompt template!

Now, try it out!

Draft a 300-word report on the criminality issues associated with drug desistance.

The screenshot displays the h2oGPTe interface. On the left, a chat window shows a user prompt: "Draft a 300-word report on the criminality issues associated with drug desistance." The H2O AI response follows, summarizing key points from a document. On the right, the "New chat" configuration panel is open, showing the "Collection" tab. The "Collection to use" dropdown is set to "Drug Use Desistance". Below this, a yellow button reads "Apply current settings as collection defaults". The "Description" field contains a text snippet about UK drug user estimates. The "Documents" section shows a table with one document: "Drug_use_desistance.pdf", which is marked as "Indexed".

Chat History:

You: Draft a 300-word report on the criminality issues associated with drug desistance.

H2O: According to the information provided, the article discusses several key points regarding the criminality issues associated with drug desistance:

1. The majority of people who use drugs do not go on to become drug dependent, with only around 1% of the UK population being considered problematic drug users. This suggests that natural desistance from drug use is the norm rather than the exception.
2. The article examines different theories for understanding drug dependence and desistance. It critiques the view that drug addiction involves a loss of individual autonomy, arguing that drug use can be better explained through the lens of matching theory, which sees it as a continual choice between drug use and other alternatives based on immediate rewards.
3. The article notes that current government policy, as exemplified by the Drug Intervention Programme and Drug Treatment and Testing Orders, is based on the premise that intervention and coercive treatment are necessary to facilitate desistance, particularly among problematic drug users involved in criminality.

New chat Configuration:

Collection: Drug Use Desistance

Description: This text is a table showing UK estimates of drug users, including the number of users, the percentage of the population, the drug definition, and the period when the drug was consumed.

Documents:

Id	Name	Status
584cfe2f-2d85-4ff7...	Drug_use_desistance.pdf	Indexed

Hint: Drag and drop files to add them to this collection!

Agent Approach



Multi-Model Integration

Combines standard models for audio-visual understanding and reasoning, paired with Sonnet 3.5 for natural language processing and reasoning tasks.

Minimalist Approach

Uses raw model without additional fine tuning

Single agent with minimal scaffolding, avoids specialized methods or complex orchestration pipelines

Code First Agent

CodeAct (Code + Act) framework extends code generation and execution as the primary form of action

Agent Analysis

Example Workflow



Prompt

Analyze the stock price of Tesla



Code Execution

Load stock price from yfinance in a python notebook



Error Handling

Error loading yfinance, switching to alphavantage api



Agent tool: Websearch

Execute google_search.py with the modified prompt



Code Execution

Utilize BeautifulSoup to scrape webpages for additional details

```
# filename: tesla_stock_analysis.py
# execution: true

import yfinance as yf
import pandas as pd
import matplotlib.pyplot as plt
from datetime import datetime, timedelta

# Get Tesla stock data for the last year
tesla = yf.Ticker("TSLA")
end_date = datetime.now()
start_date = end_date - timedelta(days=365)
tesla_data = tesla.history(start=start_date, end=end_date)

# Create a basic price analysis plot
plt.figure(figsize=(12, 6))
```

Coder Agent

I see the Alpha Vantage demo API key has limitations. Let me try using Google search to get recent Tesla stock information and analyze it:

```
# filename: search_tesla_stock.sh
# execution: true
python /workspace/openai_server/agent_tools/google_search.py --query "Tesla TSLA stock price current analysis" --type news
```

```
# filename: get_tesla_info.py
# execution: true
import requests
from bs4 import BeautifulSoup
import pandas as pd
import matplotlib.pyplot as plt
from datetime import datetime

# Function to clean price string
def clean_price(price_str):
    return float(price_str.replace('$', '').replace(',', ''))

# Get the webpage content
url = "https://finance.yahoo.com/quote/TSLA"
headers = {'User-Agent': 'Mozilla/5.0'}
```

Use GenAI on your Documents with h2oGPTe

Summary

- **Prompting (Chat)**

- Use prompting techniques to interact with LLMs to get desired response
- Prompt templates can be created and shared

- **Data Ingestion (Collections)**

- Ingest various data types (documents, audio, images, websites)
- Supports over 65 file formats grouped into “Collections” for easy management

- **Retrieval-Augmented Generation**

- A technique used to ensure contextualised response by providing relevant documents for LLM to retrieve data from - improves model’s response accuracy

- **Configuration**

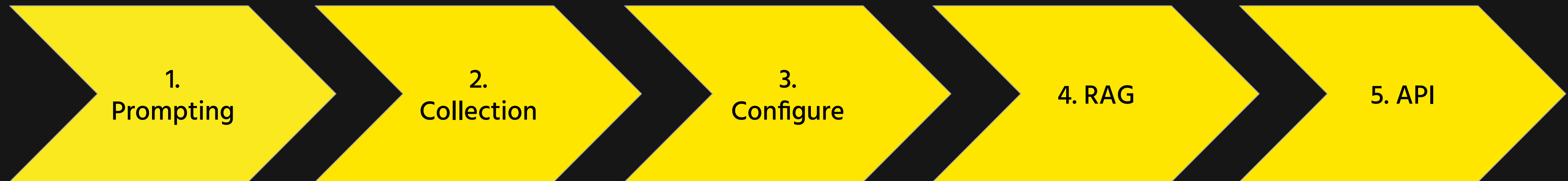
- Customise settings such as the selected LLM, RAG approach, and prompt template

- **API**

- Allow technical users to build applications around h2oGPTe via API

Thank you

Explore h2oGPTe in Action – 5 Steps



SCAN THE FOLLOWING QR CODE TO ACCESS THE PDF

Drug_use_desistance.pdf

Recommended PDF File to work with

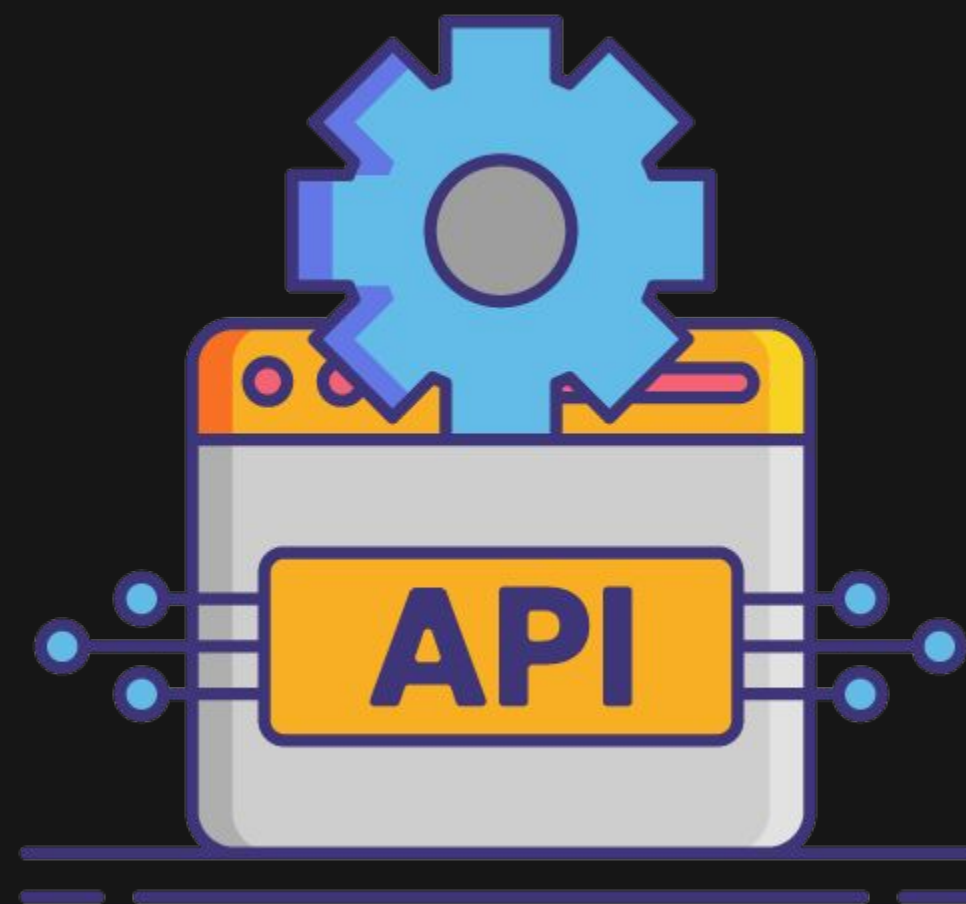


ResearchGate PDF

WHAT ARE APIs?

- **APIs act as a bridge between software applications, enabling seamless communication.**

- **Responses are provided in a structured, programmatic way for easy integration.**



- **They allow you to send requests (e.g., ask questions, retrieve insights, generate content).**
- **APIs enable embedding h2oGPTe's capabilities directly into your systems, beyond the standard interface.**

TYPES OF API KEYS

Global API Keys –

Grant full access, allowing API calls to impersonate the user.

Collection-Specific Keys –

Restrict access to a particular collection for controlled usage.